

A cascade for point-process inference

1 What the exploration showed

- **Everything breaks in the same place: near complete spatial randomness (CSR).** Across every feature set in **results/** (classical and topological):
 - Below $\tilde{\delta} \approx 0.6$ the 5-way classifiers recognise Thomas only 2–8% of the time and call 59–86% of those patterns Poisson (nested and LGCP alike: 51–82% called Poisson)
 - Parameter errors are largest near CSR and shrink as the structure strengthens: log-RMSE (mean over parameters) Thomas $\approx 0.9 \rightarrow 0.4$, LGCP $\approx 0.6 \rightarrow 0.25$ from $\tilde{\delta} < 0.5$ to $\tilde{\delta} \geq 8$ (best feature sets)

2 Design

stage	question	model	trained on
1. filter	Poisson, clustered or repulsive?	trees	all simulated patterns
2. classifiers	clustered: Thomas / nested / LGCP / <i>reject</i> repulsive: Matérn II / <i>reject</i>	trees	in-regime patterns of the branch + reject examples
3. estimators	the family’s parameters (log-MSE)	per family	that family’s in-regime patterns
Poisson	intensity	$\hat{n} = n$ (exact MLE)	—

- **Why filter:** near-CSR patterns carry no usable structure, so the filter calls them Poisson and fits $\hat{n} = n$. That is wrong as a label but fine as a fit, and every downstream model is trained only on patterns where its structure is visible
- **Reject:** patterns the filter lets through by mistake (mostly Poisson) are re-routed to Poisson by stage 2. Reject examples are Poisson patterns, the branch’s own out-of-regime patterns, and the other branch’s patterns
- **Each component is trained on its own**, from simulated patterns only; no component sees another’s predictions. They are assembled only at test time
- **The assembled pipeline is scored end to end** by a proper scoring rule on simulations from the fitted model (Section 5). This rewards a useful fit rather than a correct label, and does not reuse the summary functions the models were trained on

3 Stage 1: the filter

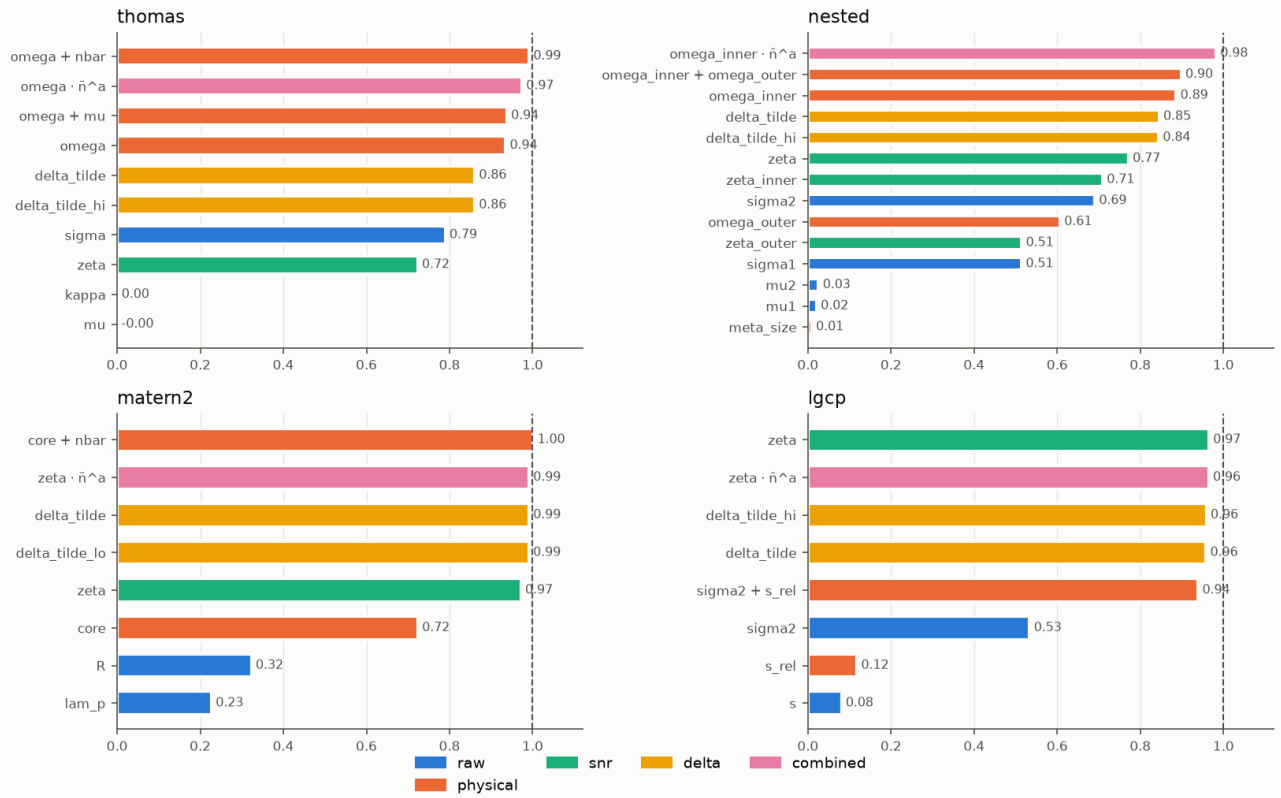
model (same 120k test patterns)	balanced accuracy
network on summary curves (L, F, G, J)	0.796
gradient-boosted trees on summary features (chosen)	0.796
logistic regression on summary features	0.794
persistence-diagram network (PersLay)	0.774

- Three very different models tie: a ceiling set by the data, not the model. Trees chosen: tied for best, CPU-only, deterministic
- **The remaining errors are the regime the filter is meant to absorb:** 77% of its errors are structured patterns called Poisson, and 90% of those have $\tilde{\delta} < 1$ (68% have $\tilde{\delta} < 0$, a departure from CSR smaller than CSR's own noise)

4 Regime analysis: where to cut

- **Question:** which function of a pattern's true parameters best predicts whether the filter sends it to its own group?
- **Candidates, per family:** raw parameters; physical ones (cluster overlap $\omega = \sigma\sqrt{\kappa}$, hard-core size, correlation range); a rough pair-count signal-to-noise ratio ζ ; and $\tilde{\delta}$
- **Score:** out-of-fold log-loss of a monotone fit, as a fraction of what a model given *all* parameters achieves (1 = the coordinate explains everything the parameters can). Fitted on the filter's held-out predictions on the *training* set, so the test set never picks the cutoff

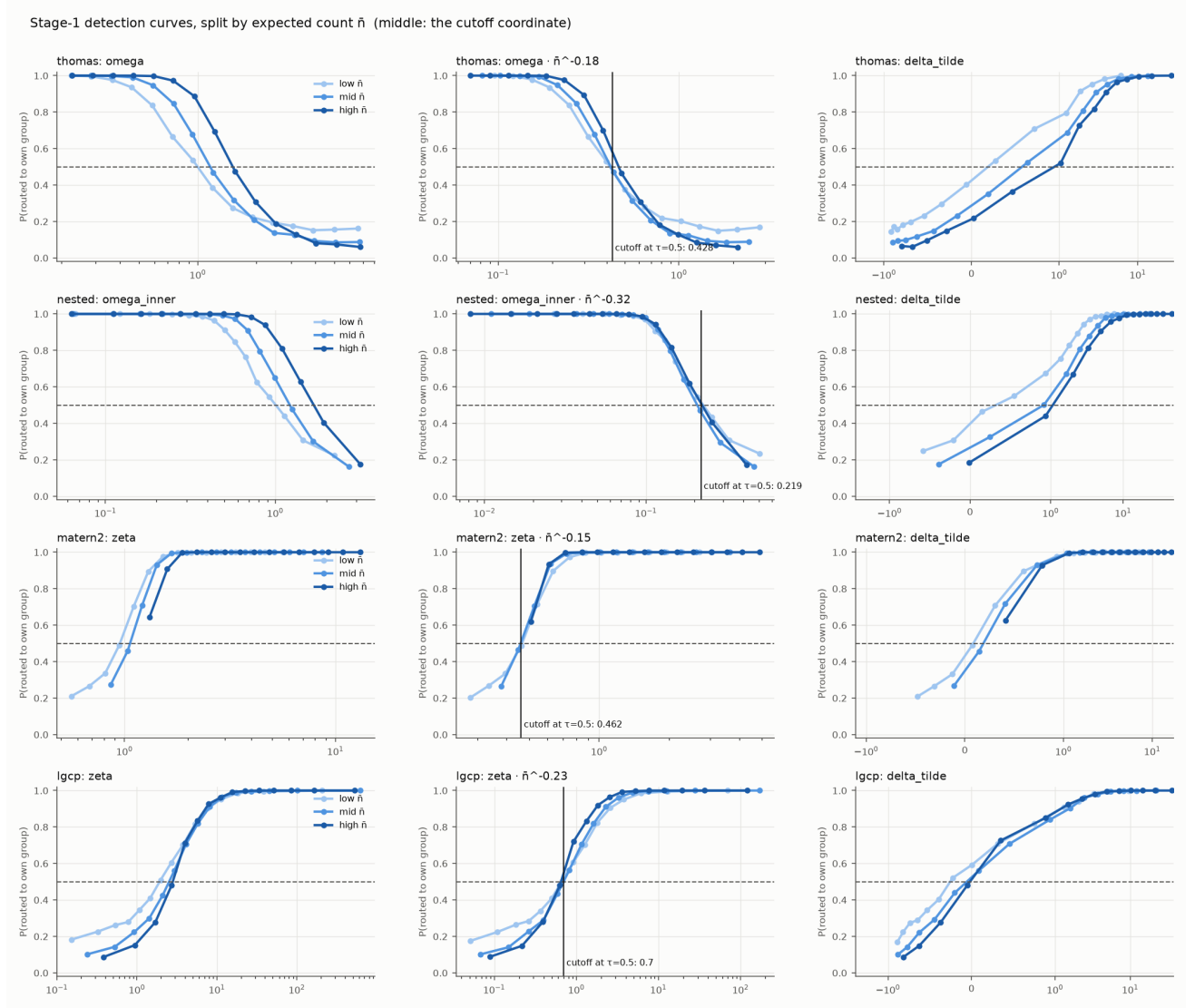
How much of stage 1's routing each coordinate explains (1 = everything θ can explain)



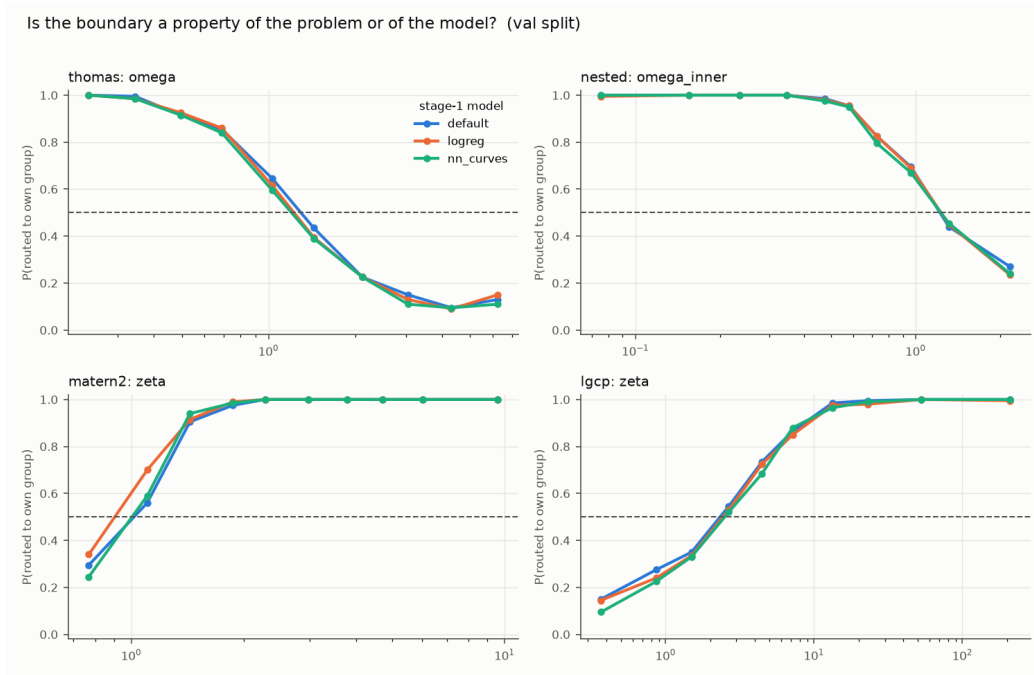
family	chosen coordinate (corrected for $\bar{n}$)	explained	$\tilde{\delta}$ explained
Thomas	cluster overlap $\omega = \sigma\sqrt{\kappa}$	0.975	0.861
nested	inner-cluster overlap	0.983	0.845
Matérn II	hard-core size $\times \bar{n}$	0.992	0.991
LGCP	field strength $\times$ range $\times \bar{n}$	0.965	0.958

- **Physical coordinates match or beat $\tilde{\delta}$.** $\tilde{\delta}$ is built from the L -function, which the filter also sees, so a physical coordinate that ties it is the cleaner choice

- **Two-dimensional:** more points make the same overlap easier to see, so each cutoff is on (coordinate, $\bar{n}$), written as one number coordinate $\times \bar{n}^a$ with a fitted



Left: raw coordinate, split by $\bar{n}$. Middle: the $\bar{n}$ -corrected coordinate the cutoff uses — the curves collapse onto one. Right: $\tilde{\delta}$.



Trees, logistic regression and the network give the same curves: the cutoff is a property of the problem, not of the filter.

From curve to training data

- A pattern is **in regime** at threshold τ if the filter sends patterns like it to their own group with probability $\geq \tau$
- $\tau \in \{0, 0.3, 0.5, 0.7, 0.9\}$; $\tau = 0$ is no filtering, the baseline the filter must beat
- Higher τ = cleaner but fewer training patterns (at $\tau = 0.5$: Thomas keeps 49%, nested 83%, Matérn II 88%, LGCP 66%)

5 Scoring the whole pipeline

- **Per test pattern x** , simulate three models and score each against x (lower is better):
 - **fit**: the pipeline's $(\widehat{\text{family}}, \hat{\theta})$
 - **truth**: the true family at the true θ — the floor
 - **CSR**: Poisson with $\bar{n} = n$ — no structure
- **Primary score: kernel score on local configurations.** Around each point, take its neighbours within 1.5 mean spacings; compare the distribution of these local configurations in x with that in the model's simulations, through a kernel. Strictly proper: the true model minimises it in expectation
- **Secondary score: Dawid–Sebastiani** on 14 geometric statistics (Voronoi cells, nearest neighbours, Delaunay edges), no persistent homology
- **Why not the obvious Wasserstein distance?** Two draws of the same clustered process do not line up point by point, so $W(x, \text{simulation})$ is large even for the true model. Tested: it could not tell the truth from CSR
- **Summaries**, pooled over ~ 300 test patterns stratified by $\tilde{\delta}$ (never read single patterns):

$$\text{regret} = S(\text{fit}) - S(\text{truth}), \quad \text{gain} = S(\text{CSR}) - S(\text{truth}), \quad \text{skill} = 1 - \frac{\sum \text{regret}}{\sum \text{gain}}$$

skill = 1: as good as the truth; 0: no better than CSR

- **Power check (truth vs CSR):** at $\tilde{\delta} > 4$ the kernel score prefers the truth on 93–100% of patterns. Most of that power comes from higher-order structure, not the pair correlation the models were trained on
- **“Might as well be Poisson” becomes testable:** on structured patterns the pipeline sends to Poisson, the fit *is* CSR, so the claim is simply gain ≈ 0

6 Results at each stage

Stage 2: classifiers, on the patterns the filter actually sends them

τ	accuracy	clustered branch		repulsive branch	
		Poisson rejected	own rejected	Poisson rejected	own rejected
0	0.610	3%	0%	4%	0%
0.3	0.614	18%	2%	13%	1%
0.5	0.597	42%	5%	27%	3%
0.7	0.566	66%	11%	45%	5%
0.9	0.505	90%	22%	68%	10%

- τ is a dial between catching Poisson impostors and losing weak real patterns (accuracy counts a rejected real pattern as wrong)
- Training past the cutoff fixes Thomas: recall $0.33 \rightarrow 0.54$ — trained on all Thomas patterns it learns “Thomas is weakly clustered”, but the filter only passes clearly clustered ones
- Main remaining confusion is Thomas vs nested, and it is identifiability: a nested process whose outer level is smeared out *is* a Thomas process (nested called nested: 96% when the outer level is crisp, 37–46% when smeared)

Stage 3: parameter estimators

RMSE(log θ) / s.d. (lower is better)	trees	curves network	PersLay rips	PersLay DTM $k=10$
Thomas	0.71	0.70	0.75	0.72
nested	0.63	0.61	0.66	0.64
Matérn II	0.13	0.13	0.17	0.18
LGCP	0.44	0.44	0.47	0.45

- **Filtering the training data does not help here:** with reject in place, an estimator trained past the cutoff and one trained on everything are within 0.02 on the patterns that reach them. Estimators are best trained on everything
- Trees and the curves network are tied at every stage. PersLay (z-scored) is behind on every family and filtration so far (DTM $k=5, 10$ and rips; $k=15, 20$ and alpha still training), at stage 2 too (clustered accuracy 0.45–0.46 vs 0.50 for trees)

7 Overall results

skill (1 = as good as the truth, 0 = no better than CSR)				
τ	trees everywhere		mixed	
	kernel	DSS	kernel	DSS
0	1.08	1.01	1.04	1.01
0.3	1.05	1.01	1.02	1.01
0.5	1.04	1.00	1.04	1.00
0.7	1.02	1.00	1.02	1.00
0.9	1.10	1.00	0.98	1.00

300 test patterns. Mixed: trees for stages 1–2; curves network for the clustered families’ parameters, PersLay (rips) for Matérn II’s. Pooled s.e. of skill ≈ 0.05 –0.10.

τ	structured patterns sent to Poisson	their gain (kernel): $\approx 0 =$ Poisson is enough
0		-0.00000 ± 0.00008
0.3		-0.00007 ± 0.00011
0.5		-0.00013 ± 0.00011
0.7		-0.00011 ± 0.00009
0.9		-0.00001 ± 0.00011

For scale: on strongly structured patterns ($\tilde{\delta} > 4$) the gain is 0.011 ± 0.0015 , so the score has power where there is structure.

- **“Might as well be Poisson” holds:** on every structured pattern the pipeline sends to Poisson, the true model scores no better than Poisson
- **Where it fits a family, the fit is as good as the truth** as far as either score can tell (skill ≈ 1)
- **τ does not matter end to end:** rejected patterns are near-CSR, so Poisson or a weak family fit are equally good
- **The estimator choice does not matter end to end either:** networks at stage 3 score the same as trees, as the per-stage results predicted
- Nested is the only family where the fit is slightly worse than the truth in both pipelines — the Thomas/nested identifiability again
- **Caveat:** the fit is scored on the pattern it was estimated from, which flatters it (hence skill slightly > 1). Next: fit on one replicate, score on the independent second replicate of the same θ